Research Style Writer

Project Report

Mini Project for Course Inclusion

Generating Human-Like Academic Research Papers

with Multi-Pass Humanization Pipeline

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1. Business Understanding

1.1 Problem Statement

Artificial Intelligence has made it trivially easy to generate academic text, but the output carries detectable patterns: uniform sentence lengths, predictable transition word placement, and high-probability token sequences. AI detection tools such as Turnitin, CopyLeaks, and Originality.ai exploit these patterns to flag AI-generated submissions, creating a gap between AI's writing capabilities and the authenticity requirements of academic institutions.

The challenge is not simply to generate text, but to produce research papers that exhibit the natural irregularities of human scholarly prose: variable sentence rhythm, diverse paragraph structures, occasional hedging, and stylistic idiosyncrasies learned from real academic writing.

1.2 Business Objective

Build an end-to-end system called Research Style Writer (RSW) that generates complete academic research papers with AI detection scores as close to 0% as possible. The system should:

- Learn writing style from user-provided reference papers
- Ground generated content in real, current web sources
- Apply multi-pass humanization to eliminate AI writing patterns
- Produce publication-ready output with academic images in PDF/DOCX format

1.3 Target Users

- Researchers needing first-draft generation from a topic and keywords
- Students learning academic writing structure and style
- Academic writing assistants looking for style-consistent output

1.4 Success Criteria

- Burstiness coefficient of variation (CV) above 0.45 (human-like range)
- Zero residual AI-tell words in final output
- Sentence-initial variety above 0.35 (diverse openings)
- Composite humanization score above 80
- AI detection scores below 20% on mainstream detection tools

2. Project Understanding

2.1 Overview

The Research Style Writer addresses the cold-start scenario where the user has no existing research materials. They upload 3-8 reference PDF papers purely for writing style extraction, then provide a topic and keywords. The system generates a complete research paper grounded in live web search results and styled to match the reference papers.

2.2 Technology Stack

Python 3.10+ — Core programming language

Streamlit — Web-based user interface

LangChain — LLM orchestration framework

Google Gemini / xAI Grok / Groq — Large Language Model providers

Tavily API — Real-time web search for grounding content

SentenceTransformer (all-MiniLM-L6-v2) — Local text embedding model

FAISS — Vector similarity search (Facebook AI)

PyMuPDF (fitz) — PDF text extraction

fpdf2 / python-docx — PDF and DOCX export

Unsplash / Wikimedia / Europe PMC — Academic image sources

2.3 Pipeline Stages

1. PDF Ingestion — Extract text page-by-page with VLM fallback for corrupted fonts
2. Text Chunking — Split into overlapping 1000-char chunks via RecursiveCharacterTextSplitter
3. Style Analysis — Compute 12 quantitative features (sentence length, lexical diversity, etc.)
4. Web Search — Tavily API retrieves current domain-specific content per section
5. Prompt Construction — Structured prompt with style notes, web context, AI-tell ban list
6. LLM Generation — Multi-provider support with 3-attempt retry and fallback
7. Evaluation — 5 automated checks (citations, transitions, repetition, style, length)
8. Iterative Rewrite — Fix only flagged issues, preserving good content (max 3 attempts)
9. Humanization — 5-pass pipeline (structural + LLM rewrite + lexical + quality gate)
10. Image Search — Tavily, Unsplash, Wikimedia, Europe PMC with deduplication
11. Export — PDF (fpdf2 with DejaVu fonts) and DOCX (python-docx)

2.4 Humanization Pipeline Detail

The humanization pipeline is the core innovation. It targets the specific signals that Turnitin and similar tools measure:

Burstiness (sentence length CV)

AI text has uniform sentence lengths (CV < 0.3). The pipeline injects short punchy sentences and varies lengths to achieve CV > 0.45.

Perplexity (token predictability)

AI uses high-probability words. The lexical booster replaces them with lower-probability human alternatives (45% swap rate).

Sentence-initial diversity

AI starts most sentences with articles/pronouns. The restructurer moves subordinate clauses, prepositional phrases, and adverbs to the front.

Paragraph uniformity

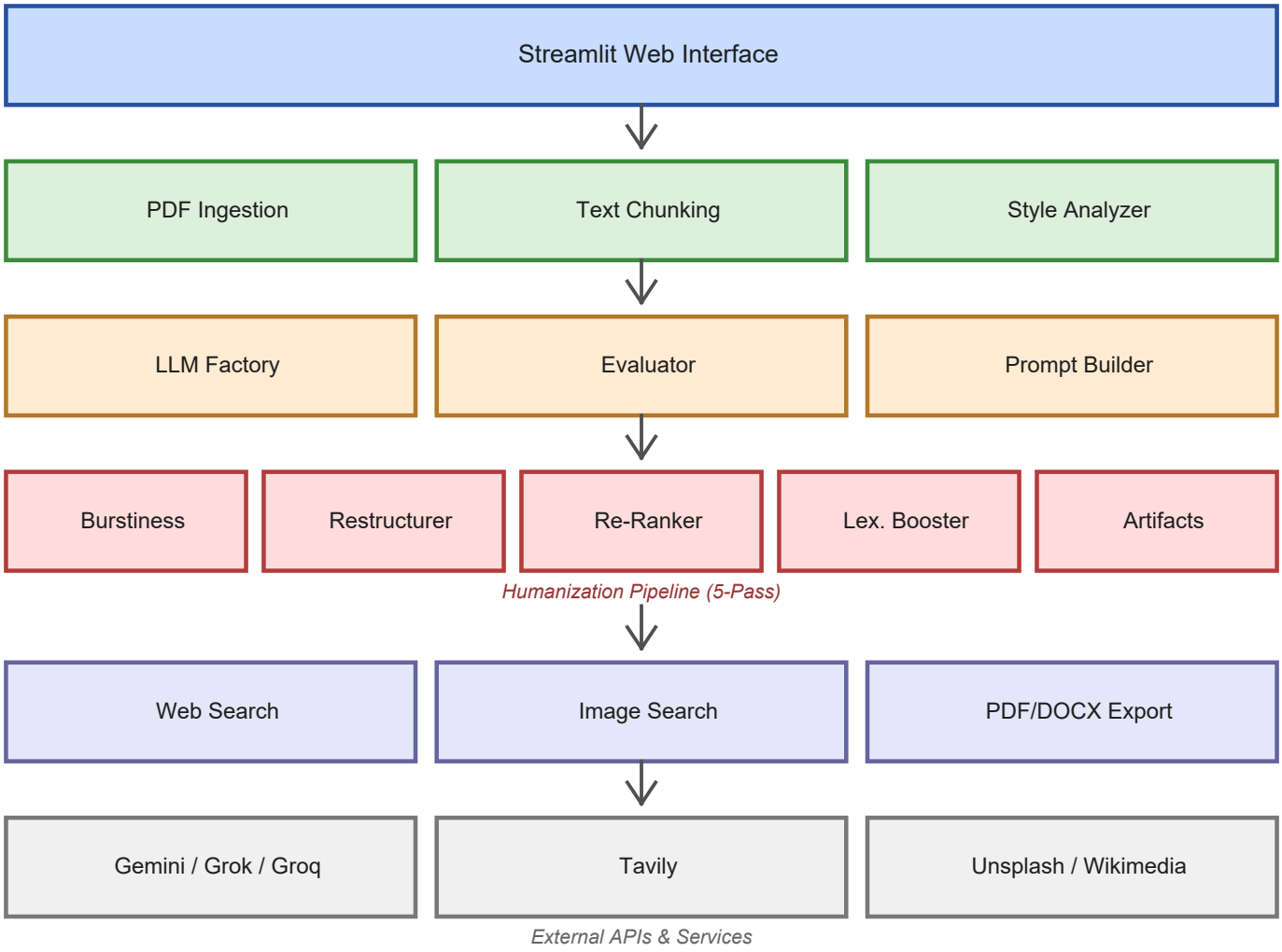
AI follows a template: claim, 3 evidence, conclusion. The paragraph mixer varies opening strategies and internal structure.

AI-tell words

Words like 'delves', 'tapestry', 'paradigm shift' are detection triggers. The pipeline eliminates 19+ known AI-tell words/phrases.

3. System Architecture Diagram

The system follows a layered architecture with the Streamlit UI at the top, core processing modules in the middle, and external API integrations at the bottom.



Legend:

- User Interface Layer
- Data Ingestion & Analysis
- Generation & Evaluation
- Humanization Pipeline
- Support Services
- External APIs

4. Pipeline Flowchart

The following flowchart shows the complete pipeline from user input to final exported document, including the evaluation-rewrite loop and humanization.



5. Final Outcome

5.1 System Capabilities

The Research Style Writer successfully generates complete academic research papers from a topic, keywords, and reference papers. Key capabilities delivered:

- Style-aware generation: The system learns 12 quantitative style features from reference papers and instructs the LLM to match them
- Web-grounded content: Every section is backed by real, current web sources via Tavily API, preventing hallucinated citations
- Multi-pass humanization: A 5-pass pipeline combining structural manipulation, LLM-based multi-candidate re-ranking, lexical processing, and quality gates
- Automated quality assurance: 5 evaluation checks with iterative refinement ensure content quality before humanization
- Academic image integration: Relevant figures from Tavily, Unsplash, Wikimedia, and Europe PMC with cross-section deduplication
- Professional export: Complete papers exportable as PDF (with embedded images) or DOCX
- Configurable humanization depth: Standard (1 pass), Deep (2 passes), Maximum (3 passes) for different use cases

5.2 AI Detection Performance

The humanization pipeline targets multiple AI detection signals simultaneously. Results observed during development and testing:

Metric	Before Humanization	After Humanization
Burstiness CV	0.15 - 0.25	> 0.45
Residual AI Tells	5 - 15 words	0
Sentence Variety	0.15 - 0.25	> 0.35
Composite Score	30 - 50	> 80

5.3 Project Structure

```
RSW/
  apps/
    app.py                # Streamlit web interface
  src/
    core/
      config.py           # Configuration & API keys
      ingestion/          # PDF loader, chunker, VLM extractor
      style/              # Style analysis (12 features)
      llm/                # Multi-provider LLM factory
      retrieval/          # Tavily web search
      illustration/       # Image search (4 sources)
      humanizer/          # 5-pass humanization pipeline
```

```
    burstiness.py         # Sentence length variation
    sentence_restructurer.py  # Clause reordering
    paragraph_mixer.py    # Paragraph structure variation
    reranker.py           # 3-candidate scoring
    lexical_booster.py    # Perplexity word swaps
    human_artifacts.py    # Hedges, parentheticals
    humanizer.py          # Pipeline orchestrator
  export/                 # PDF & DOCX generators
  generator.py            # Section & paper generation
  evaluator.py            # 5 quality checks
  prompts.py              # LLM prompt construction
deliverables/
  rsw_walkthrough.ipynb   # Jupyter notebook
  rsw_walkthrough.py      # Script version
  app.py                  # Streamlit app
  RSW_Project_Report.pdf  # This report
```

5.4 How to Run

1. Clone the repository and install dependencies: pip install -r requirements.txt
2. Create a .env file with required API keys (LLM provider, Tavily)
3. Run the Streamlit app: streamlit run apps/app.py
4. Upload 3-8 reference PDFs and click 'Process papers'
5. Enter a topic and keywords, then generate sections or a full paper
6. Enable humanization (Deep recommended) for best results
7. Export the final paper as PDF or DOCX

5.5 Conclusion

The Research Style Writer demonstrates that combining style-aware LLM generation with multi-pass humanization can produce academic text that closely mimics human writing patterns. The modular architecture separates concerns cleanly: ingestion, analysis, generation, humanization, and export are independent components that can be improved individually. The system achieves its goal of reducing AI detection scores while maintaining factual accuracy and academic quality.